



Research Article

Crucial role played by CK8⁺ cells in mediating alveolar injury remodeling for patients with COVID-19Yufeng Li^{a,c}, Hengrui Hu^a, Jia Liu^a, Longda Ma^b, Xi Wang^a, Liang Liu^b, Qian Liu^b, Liang Ren^b, Jiang Li^a, Fei Deng^a, Zhihong Hu^a, Yiwu Zhou^{b,*}, Manli Wang^{a,*}^a State Key Laboratory of Virology, Wuhan Institute of Virology, Center for Biosafety Mega-Science, Chinese Academy of Sciences, Wuhan, 430071, China^b Department of Forensic Medicine, Tongji Medical College of Huazhong University of Science and Technology, Wuhan, 430010, China^c University of the Chinese Academy of Sciences, Beijing, 100049, China

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ABSTRACT

The high risk of SARS-CoV-2 infection and reinfection and the occurrence of post-acute pulmonary sequelae have highlighted the importance of understanding the mechanism underlying lung repair after injury. To address this concern, comparative and systematic analyses of SARS-CoV-2 infection in COVID-19 patients and animals were conducted. In the lungs of nine patients who died of COVID-19 and one recovered from COVID-19 but died of unrelated disease in early 2020, damage-related transient progenitor (DATP) cells expressing CK8 marker proliferated significantly. These CK8⁺ DATP cells were derived from bronchial CK5⁺ basal cells. However, they showed different cell fate toward differentiation into type I alveolar cells in the deceased and convalescent patients, respectively. By using a self-limiting hamster infection model mimicking the dynamic process of lung injury remodeling in mild COVID-19 patients, the accumulation and regression of CK8⁺ cell marker were found to be closely associated with the disease course. Finally, we examined the autopsied lungs of two patients who died of infection by the recent Omicron variant and found that they only exhibited mild pathological injury with no CK8⁺ cell proliferation. These results indicate a clear pulmonary cell remodeling route and suggest that CK8⁺ DATP cells play a primary role in mediating alveolar remodeling, highlighting their potential applications as diagnostic markers and therapeutic targets.

1. Introduction

Since the end of 2019, SARS-CoV-2 has spread worldwide and has become a public health emergency of international concern. Early epidemic viral strains mainly attack the lower respiratory tract and cause respiratory distress syndrome, as evidenced by severe lung pathological injuries, including inflammatory cell infiltration, diffuse alveolar injury, and pulmonary fibrosis in autopsied patient samples (Ducloyer et al., 2020; Fox et al., 2020; Gu et al., 2005; Liu et al., 2021; Schwensen et al., 2020; Tian et al., 2020; Xu et al., 2020; Zhao et al., 2021). The currently prevalent Omicron variants mainly replicate in the upper respiratory tract and generally cause milder lung diseases than earlier epidemic strains (Hui et al., 2022; Shuai et al., 2022); however, severe and lethal cases have also been reported (Goussard et al., 2022; Märkl et al., 2023). While most people can recover from SARS-CoV-2 infection, some suffer long-term lung sequelae (long COVID-19), whose symptoms include shortness of breath and coughing (Davis et al., 2023). The diffusion

capacity of the lungs for carbon monoxide anomalies has been reported to occur in 16.4% of convalescent patients, and over 50% of previously hospitalized survivors show abnormalities on computed tomography (CT) (Blanco et al., 2021; Solomon et al., 2021). This raises public concern regarding the post-acute pulmonary sequelae of COVID-19, and the lung injury repair mechanism is poorly understood.

The lungs are exposed to versatile external environments and microorganisms and face the risk of various injuries, hence their high repair capability. A small population of pluripotent stem cells/progenitor cells, such as club and basal cells, and bronchoalveolar stem cells (BASCs), self-expand and differentiate to mediate alveolar regeneration after severe alveolar injury (Kathiriya et al., 2020; Kim et al., 2005; Kumar et al., 2011; Liu et al., 2019; Rawlins et al., 2009; Vaughan et al., 2015). For example, in influenza infections, p63⁺CK5⁺ distal airway stem cells are essential for lung regeneration and can differentiate into type I and II pneumocytes (AT I and AT II) (Kumar et al., 2011; Zuo et al., 2015). AT II cells self-renew and differentiate into AT I cells to mediate alveolar

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regeneration (Barkauskas et al., 2013; Basil et al., 2020). In SARS-CoV-2 infection, CK5⁺ basal cells proliferate extensively in the trachea and relatively large airways, while AT II cells proliferate in severely damaged alveoli for regeneration (Fang et al., 2020).

In studies on alveolar regeneration after injury and lung fibrosis, a cluster of damage-related transient progenitor (DATP) cells (cell markers: CK8, CLDN4, and CDKN1A) proliferate after alveolar injury and are associated with alveolar regeneration (Choi et al., 2020; Redente et al., 2021; Strunz et al., 2020). DATP cells could originate from club-, basal-(like), and AT II cells, which further differentiate into AT I cells to mediate proper alveolar remodeling (Kobayashi et al., 2020; Ray et al., 2016; Strunz et al., 2020). On the contrary, the deficiency of CK8⁺ cell differentiation into AT I cells was often associated with lung diseases. For example, in idiopathic pulmonary fibrosis, obstruction in differentiation of accumulated DATP cells into AT I cells promotes the expansion of fibrotic pathological fibroblasts (Choi et al., 2020; Kobayashi et al., 2020). Single-cell sequencing of lung autopsy samples of deceased COVID-19 patients with severe alveolar injury also detected DATP cell (CK8 and CLDN4) accumulation, which failed to undergo full transition into AT I cells, suggesting impaired epithelial differentiation and lung regenerative processes (Melms et al., 2021; Wang et al., 2021).

The limitation of human sample sources complicates observation of the different injury-repair stages of patients with COVID-19. Animal models may serve as surrogates for studying different disease courses. Mouse models of SARS-CoV-2 infection, either AdV-hACE2 transduction mice or ACE2 transgenic mice, simulate human pulmonary pathological symptoms, mainly with respect to inflammatory cell infiltration. In contrast, golden Syrian hamsters can be infected with SARS-CoV-2 naturally and rapidly recover, and their pulmonary lesions exhibit diffuse alveolar damage, AT II hyperplasia, and an inflammatory response (Feng et al., 2022; Mulka et al., 2022). Therefore, the hamster infection model has certain advantages over mice in terms of studying the dynamic process of lung injury remodeling in COVID-19.

In this study, we used autopsied lung samples from nine patients who died of COVID-19 and one patient recovered from COVID-19 but died of unrelated disease (all infected with the SARS-CoV-2 original strain in 2020) to analyze the repair status of lung injury post-infection. Using a hamster infection model, we dynamically analyzed the pathology and repair of the lungs after SARS-CoV-2 infection. Numerous CK8⁺ DATP cells were identified in both the human patients and hamster infection

model, which derived from CK5⁺ basal-(like) cells. In COVID-19 convalescent patient or infected hamsters, CK8⁺ cells could differentiate into RAGE⁺ AT I cells; however, this was seldom detected in autopsied samples of patients who died of COVID-19. Furthermore, we detected the lung samples from two patients infected with the recent Omicron pandemic (at the end of 2022), with no obvious proliferation of CK8⁺ cells. In summary, comparative and systematic analyses showed that CK8⁺ cells play a crucial role in the regeneration of injured lungs for patients with COVID-19 and highlighted their potential in disease diagnosis and therapy.

2. Materials and methods

2.1. COVID-19 patients, healthy donors, and clinical data

We used autopsy lung tissue from nine patients who died of COVID-19 and one patient recovered from COVID-19 but died of unrelated disease with the original strain infection at the beginning of 2020, and two deceased patients infected with the Omicron BA.5 strain at the end of 2022/early 2023 were provided by Wuhan Jinyintan Hospital and Tongji Medical College of Huazhong University of Science and Technology (background information is provided in Table 1). In addition, six healthy donors provided by Tongji Medical College of Huazhong University of Science and Technology were included in the study (background information is provided in Table 1, healthy donor (HD) 1 was used for pulmonary pathological examination, HD 2–6 were used for RNA-seq). Lung tissues were collected according to the regulations issued by the National Health Commission of China and were approved by the Ethics Committee of the hospital. Written informed consent was obtained from the patient's family before autopsy.

2.2. Hamster infection experiments

SARS-CoV-2 original strain (2019-nCoV-WIV04 strain; NVRC: IVCAS 6.7512) was propagated in Vero E6 cells, the virus stock was stored at –80 °C and titrated by an end-point dilution assay as described in a previous study (Zhou et al., 2020). Five- to six-week-old Syrian golden male hamsters infected with 1 × 10⁵ TCID₅₀ of SARS-CoV-2 original strain per hamster through intranasal administration (volume: 50 μL) or 50 μL PBS as a control. Body weight was monitored daily and animals in

Table 1
Clinical data and laboratory tests of the COVID-19 patients and healthy donors.

Patients No	Gender	Virus strain	Age (years)	Death time (days after onset)	Pulmonary lesion				Pulmonary virus detection	Cause of death
					①	②	③	④		
Case 1	F	Original strain	66	17	4	2	3	0	+	COVID-19
Case 2	M		72	32	4	2	3	1	+	COVID-19
Case 3	F		53	34	4	2	2	2	+	COVID-19
Case 4	M		85	30	4	2	3	0	+	COVID-19
Case 5	F		86	24	4	2	3	1	+	COVID-19
Case 6	M		62	31	4	2	3	0	+	COVID-19
Case 7	M		51	28	4	2	3	2	+	COVID-19
Case 8	M		78	27	4	1	3	0	+	COVID-19
Case 9	F		67	22	4	1	3	1	+	COVID-19
Case 10	M		72	47	2	0	1	0	+	Liver Cirrhosis in Combination with Upper Digestive Tract Hemorrhage
Case 11	M	Omicron strain	16	2	0	4	1	2	+	COVID-19
Case 12	M		47	15	0	3	1	1	–	COVID-19
HD 1	F	/	21	/	0	0	0	0	–	Carbon monoxide poisoning
HD 2	F		21	/	0	0	0	0	–	Traffic accident
HD 3	M		32	/	0	0	0	0	–	Coronary artery myocardial bridge
HD 4	M		27	/	0	0	0	0	–	Coronary heart disease combined with high heart disease
HD 5	M		55	/	0	0	0	0	–	Brain damage
HD 6	M		57	/	0	0	0	0	–	Coronary heart disease

①: pulmonary fibrosis score; ②: pulmonary edema score; ③: Inflammatory cell infiltration score; ④: Pulmonary hemorrhage score. The score indicates the lesion area accounts for the percentage of all alveoli. 1 score: 1%–25%; 2 score: 26%–50%; 3 score: 51%–75%; 4 score: 76%–100%.

the infection group were euthanized at 3, 6, 9, and 11 dpi (n = 5, 4, 5, and 4, respectively). In the control group, samples were collected at 6 dpi (n = 5).

2.3. Quantitative real-time polymerase chain reaction

Total tissue RNA was extracted using the TRIzol reagent. Then 1 µg RNA was reverse transcribed to cDNA using the PrimeScript RT Reagent Kit with gDNA Eraser (Takara, Cat no. RR047A). For throat swabs, RNA was extracted using a MiniBEST Viral RNA/DNA Extraction Kit (Takara, CAT No. 9766) and then 3 µL RNA was reverse transcribed to cDNA. Afterwards, 1 µL cDNA was subjected to quantitative polymerase chain reaction (PCR) using TB Green Premix Ex Taq II (RR820A, Takara) and primers RBD-qF: 5'-CAATGGTTTAACAGGCACAGG-3' and RBD-qR: 5'-CTCAAGTGTCTGTGGATCAGC-3'.

2.4. Histochemical staining

Lung samples were paraffin embedded after 4% paraformaldehyde fixing and dehydrating, then cut into 3 µm sections for staining. For hematoxylin-eosin (H&E) staining, the slides were deparaffinized by soaking in xylene twice for 15 min, and then rehydrated by soaking in 100%, 100%, 85%, and 75% ethanol for 5 min successively. The slides were then stained with hematoxylin for 5 min before differentiation for 30 s in 0.5% hydrochloric acid ethanol and blue for 30 s in 0.2% ammonia water and eosin for 10 min. After dehydration, the slides were sealed with neutral gum. Finally, the slides were scanned using a Panoramic MIDI system (3DHISTECH; Thermo Fisher Scientific).

2.5. Masson's trichrome staining

The slides were cut into 3 µm and deparaffinized, then stained using Masson's trichrome staining kit (G1006, Servicebio). Afterwards, the slides were sealed with neutral gum and dehydrated, then scanned using a Panoramic MIDI system (3DHISTECH, Thermo Fisher Scientific).

2.6. Immunofluorescence staining

First, 3 µm paraffin sections were deparaffinized and antigens were retrieved using citric acid antigen repair solution (G1202, Servicebio). Sections for IFA staining were treated with 0.1% Sudan Black B (Sigma-Aldrich) for 30 min to reduce autofluorescence after deparaffinization. The sections were blocked with 5% BSA for 1 h at room temperature, followed by incubation with primary antibodies diluted appropriately (Supplementary Table S1) at 4 °C overnight. The slides were washed thrice with PBS and then incubated with Alexa Fluor 488/555-conjugated goat anti-rabbit or mouse IgG (1:500; Abcam) as secondary antibodies for 2 h at room temperature and then stained with DAPI (Biyuntian) after washing with PBS three times. After mounting with an anti-fluorescence quenching sealing agent (G1401, Servicebio), the slides were scanned using a Panoramic MIDI system (3DHISTECH, Thermo Fisher Scientific).

2.7. Multicolor immunofluorescence staining assay

After deparaffinization, antigen retrieval and treatment with 0.1% Sudan Black B (Sigma-Aldrich), the slides were stained following the protocol for TSA-dendron fluorophores (Opal 7-color Manual IHC kit, PerkinElmer, NEL811001kt). Primary antibodies were used at the appropriate dilutions after the preliminary experiment (Supplementary Table S1). The experimental steps have been described in detail in a previous study (Liu et al., 2021). Finally, the slides were scanned using the PerkinElmer VECTRA3 automatic quantitative analysis system.

2.8. RNA-seq and data analysis of autopsied lung tissues of COVID-19 patients

Autopsied lung tissues (deceased COVID-19 patients: n = 9, convalescent patient with COVID-19: n = 1) from patients with COVID-19 and five lung tissues from healthy donors were lysed in TRIzol. Total RNA was extracted according to the manufacturer's instructions and used to construct RNA-seq libraries following the standard process for mRNA sequencing by the Beijing Genomics Institute (BGI). Raw data were filtered using SOAPnuke (v1.5.2). The expression levels of genes were first quantified in salmon (v1.9.0) using the HomosapiensGRCh38_108.gtf genome sequences as an index. Differential gene expression and KEGG pathway enrichment analyses were performed using DESeq2 (v1.38.3) and ClusterProfiler (v4.6.2). Differentially expressed genes were selected based on padj < 0.05, log₂FoldChange > 1, and enriched pathways based on p-value Cutoff < 0.05. Graphs and analyses were generated using R (v4.2.3) and ggplot2 (v3.4.2) with custom scripts.

2.9. RNA-seq and data analysis of SARS-CoV-2 infected lungs of hamsters

The hamster lung tissues of the mock, 3 dpi, 6 dpi, 9 dpi and 11 dpi groups (n = 3, 2, 3, 2, and 2, respectively) were lysed in TRIzol. Total RNA was extracted and RNA-seq libraries were constructed as described above. Raw data were filtered using SOAPnuke (v1.5.2). The clean data were mapped to the assembled unique genes using Bowtie2 (v. 2.2.5) with the reference genome of *Mesocricetus auratus* as an index. Gene expression levels were calculated using RSEM software (v1.3.1). Differential expression analysis was performed using the DESeq2 (v1.4.5) with Q value ≤ 0.05 or FDR < 0.001. KEGG pathway enrichment analysis was performed using Phyper software based on Hypergeometric test. The significant levels of pathways were corrected using Q value with a rigorous threshold (Q value ≤ 0.05). Data analyses were provided by BGI (<https://www.bgi.com/global/service/dr-tom>). The graphs were generated using R (v4.2.3) and ggplot2 (v3.4.2) by custom scripts.

3. Results

3.1. Pulmonary fibrosis and extensive extracellular matrix induction in patients infected with the original SARS-CoV-2 strain

Lung tissue samples autopsied from nine patients who died of COVID-19 (Cases 1–9 in Table 1) during the early months of the pandemic in 2020 were subjected to pathological analysis. In addition, a convalescent patient with COVID-19 (Case 10 in Table 1) who had been discharged from the hospital for approximately one month but suddenly died of cirrhosis-induced hemorrhage was included and was autopsied for tissue samples collection within 24 h after death. Compared with healthy donors, obvious injury including pulmonary fibrosis, pulmonary edema, inflammatory cell infiltration and pulmonary hemorrhage were observed in all the 10 cases, with the convalescent patient (case 10) showing milder lung injury than the nine deceased patients (Fig. 1A and Supplementary Fig. S1). Masson's trichrome staining revealed significant fibrosis in their lungs (Fig. 1A and Supplementary Fig. S1). Immunofluorescence (IF) staining of alpha-smooth muscle actin (α-SMA, a marker of smooth muscle cells) and collagen III detected extensive myofibroblast cell proliferation and collagen fiber exudation in both deceased patients and convalescent patient, with the situation in the latter was milder (Fig. 1A and Supplementary Fig. S1). A large number of proliferated cells were found in the alveoli, predominantly CK8⁺, CK5⁺, and Ki67⁺ cells, accompanied by a significant reduction in RAGE⁺ AT I cells (Fig. 1B, upper panel) were detected in the deceased patients. In the convalescent patient, CK8⁺ and Ki67⁺ cell proliferation, but a few CK5⁺ cells were detected in the alveoli, and the RAGE⁺ AT I cells were equivalent to those

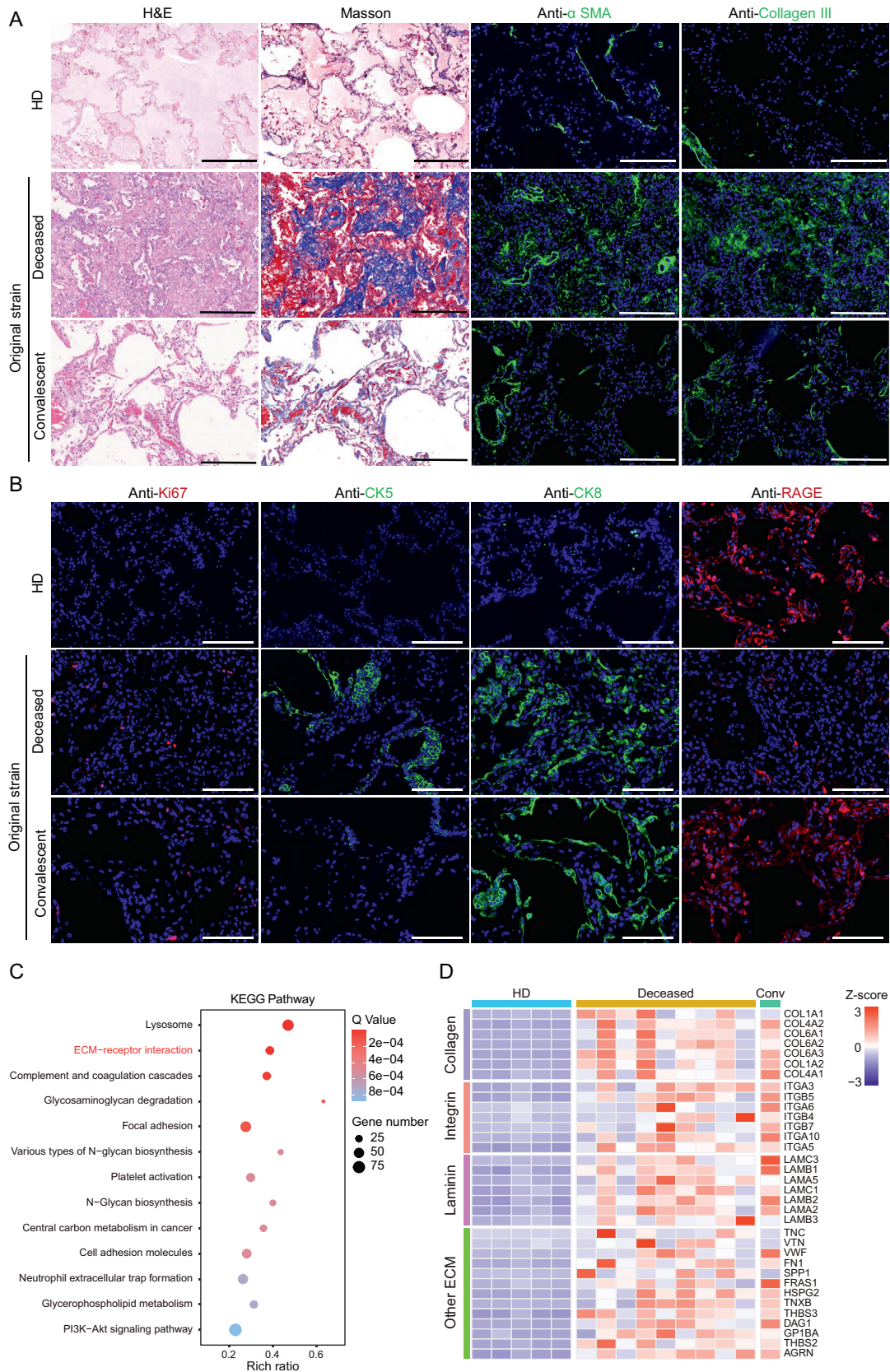


Fig. 1. Histopathological analyses of autopsied lung samples from deceased patients infected with the original COVID-19 strain. **A** Hematoxylin and eosin (H&E) staining of lung samples from patients with COVID-19 (deceased and convalescent cases) and HD (healthy donors). Masson's trichrome staining (keratin and muscle fibers were red; collagens were blue; nuclei were dark brown to black) and IF staining of alpha-smooth muscle actin (α -SMA, green) and collagen III (green). **B** IF staining of Ki67 (green), CK5 (green), CK8 (green), and RAGE (red). **C, D** Transcriptome analyses of lung tissues from deceased/convalescent COVID-19 patients and healthy donors. Conv: convalescent. Bars: 200 μ m.

of healthy donors (Fig. 1B, lower panel), suggesting that the lung injury in this patient was undergoing repair.

Transcriptome analysis of the lung tissues all the ten patients and five healthy donors identified enriched biological processes, including lysosomes, extracellular matrix (ECM) receptor interaction, and complement and coagulation in all the ten patients (Fig. 1C). As pulmonary fibrosis is usually accompanied by ECM deposition, we focused on the analysis of EMC-related pathways and found that genes belonging to major subclades of the ECM family (Kular et al., 2014), including collagen, integrin, and laminin, were significantly upregulated in deceased patients and convalescent patient with COVID-19 compared to healthy donors (Fig. 1D).

3.2. Abnormal cell proliferation and repair failure in the lungs of the deceased COVID-19 patients

Pulmonary stem cells can mediate alveolar repair after injury, we detected the three types of lung stem-like cells using multi-color IF staining. The results showed the presence of CC10⁺ club cells and CK5⁺ basal cells in the bronchioles of healthy donors, while CK8⁺ cells were mainly found in bronchioles, co-localized with CK5 in basal cells or with CC10 in club cells (Fig. 2A, upper panel), which is consistent with previous findings (Parekh et al., 2020). In the deceased COVID-19 patients, CK5⁺ CK8⁺ basal cells, but not CC10⁺ club cells, proliferated in the alveoli (Fig. 2A, middle panel). However, a portion of the proliferating CK8⁺ cells did not express CK5 (Fig. 2A, middle panel). Certain CK5⁺/CK8⁺ cells expressed proliferation marker Ki67 (Supplementary Fig. S2A and B), suggesting they were in proliferation.

The proliferative CK8⁺ DATP cells are reported mediating remodeling of alveolar epithelial cells. To detect the possible relationship between CK8⁺ and alveolar cells, multi-color IF staining of SPC (AT II cells), RAGE (AT I cells) and CK8 (DATP cells) was conducted to test the assumption that CK8⁺ cells differentiate into alveolar epithelial cells. The results showed that the number of AT II cells increased, whereas the number of AT I cells substantially decreased compared to that in healthy controls. CK8 markers co-localized with SPC in AT II cells but were rarely found in RAGE⁺ AT I cell in deceased COVID-19 patients (Fig. 2B).

3.3. Partial pulmonary remodeling status of a convalescent COVID-19 patient in early 2020 pandemic

As well as lung injury and CK8⁺ cell proliferation was also observed in convalescent COVID-19 patient, we conducted multi-color IF staining of CC10, CK5 and CK8. The results showed few CK5⁺ cells and CC10⁺ club cells in the alveoli, which differs from the situation in deceased COVID-19 patients. However, the hyperplasia of CK8⁺ cells was similar to that observed in deceased COVID-19 patients (Fig. 2A, lower panel).

Detection of alveolar cells revealed that the number of SPC⁺ AT II cells was also higher in the convalescent COVID-19 patient than in healthy donors, as found in the nine deceased patients. RAGE⁺ AT I cell counts were significantly higher than those in deceased COVID-19 patients, almost comparable with those in healthy donors (Fig. 2B). Numerous CK8 markers were co-localized with RAGE and exhibited the typical flattened and elongated shape of AT I cells, suggesting that CK8⁺ DATP cells progressively differentiated into RAGE⁺ AT I cells, a signature of alveolar epithelial remodeling (Fig. 2B). Interestingly, a few CK8⁺ and CK5⁺ cells still expressed Ki67 in the convalescent patient (Supplementary Fig. S2A and B). These results suggest that CK8⁺ DATP cells are derived from CK5⁺ basal cells and proliferate in the alveoli of patients with COVID-19. In the convalescent patient, CK8⁺ DATP cells can differentiate into AT I cells. In contrast, only a few CK8⁺ DATP cells

differentiated into RAGE⁺ AT I cells in the deceased COVID-19 patients (Fig. 2C).

3.4. Dynamic lung injury and repairing in self-limiting SARS-CoV-2 infection hamster model

To mimic the dynamic lung injury repair process in COVID-19 patients, we constructed Syrian golden hamster infection model (Fig. 3A). In the course of the 11-day infection, daily monitoring showed reduced mobility and diet, and body weight loss (up to 14.6%) during 3–6 days post-infection (dpi), and then recovered gradually. Consistently, viral infection was prominent at 3 and 6 dpi (Fig. 3B–C and Supplementary Fig. S3A). And pulmonary pathology also peaked at 6 dpi, including diffuse alveolar injury, massive cells hyperplasia, inflammatory cell infiltration, and fibrin deposition, and then gradually recovered (Fig. 3D). Masson's trichrome staining and the detection of-SMA and collagen III revealed that fibrinous exudates peaked in the alveoli at 6 dpi (Fig. 3E). Based on transcriptome analysis of the lung tissues of hamsters, six enriched biological processes were identified: the proteasome, phagosome, and NOD-like receptor signaling pathways, antigen processing and presentation, osteoclast differentiation, and glutathione metabolism at 6 dpi, compared to the PBS group, which reflected an antiviral and tissue damage repair stage (Fig. 3F). We also analyzed EMC-related pathways and categorized them into two subclades based on the changing trends. A cluster of ECM was largely upregulated at 3–6 dpi and then downregulated, including Syndecan Proteoglycan 4 (SDC4), which is related to cytoskeleton rearrangement after infection, and thrombospondin 1 (THBS1), which is related to angiogenesis. In contrast, the other cluster including von Willebrand factor (VWF) and Heparan Sulfate Proteoglycan 2 (HSPG2), which are involved in the vascular endothelial barrier, was downregulated at 3–9 dpi, and increased at 11 dpi (Fig. 3G).

In the infected hamsters, alveolar injury was accompanied by the proliferation of CK8⁺ and CK5⁺ cells (Fig. 4A and Supplementary Fig. S3B). CK5⁺ and CK8⁺ cell markers showed perfect co-localization, which was enriched at 6 dpi and gradually decreased there after (Fig. 4A and Supplementary Fig. S4A). The number of SPC⁺ AT II cells increased, but that of RAGE⁺ AT I cells decreased at 6 dpi, suggesting severe alveolar damage (Fig. 4B and Supplementary Fig. S3B). At 11 dpi, the lung structure of infected hamsters was largely restored, with only some CK8⁺ cell hyperplasia. CK8 partially co-localized with SPC or RAGE at both 6 and 11 dpi (Fig. 4B), suggesting that CK8⁺ DATP cells were involved in AT II and AT I cell differentiation. Different from the situation in patients (Supplementary Fig. S2A and B), most proliferative CK8⁺ and CK5⁺ cells expressed Ki67 (Supplementary Fig. S4B). The results showed that in SARS-CoV-2 infected hamsters, numerous CK8⁺ CK5⁺ cells proliferated and differentiated into alveolar epithelial cells to mediate lung injury repair.

3.5. Mild lung lesions in patients who died of Omicron BA.5 infection

We analyzed lung samples from two patients who died of SARS-CoV-2 Omicron infection. One was a 16-year-old male who died 3 days after disease onset, still experiencing acute fever (Case-11 in Table 1). The qPCR Ct (cycle threshold) value of the nasal swabs was 16.5 and the virus loads of tissue samples of two different pulmonary lobes were 5×10^3 or 2.7×10^4 copies/ μ g RNA (Fig. 5A and B). The other was a 50-year-old male who died 15 days after the disease onset (Case 12 in Table 1); the viral RNA in their lungs was almost cleared (Fig. 5A and B). IF staining also confirmed the results of qPCR detection (Fig. 5E). H&E staining showed that the lung lesions in both patients were mild, with small amounts of inflammatory cells infiltrating around the bronchioles (Fig. 5C). Masson's trichrome staining and IF staining of α -SMA and collagen III did not show collagen fibrinous exudation and myofibroblast

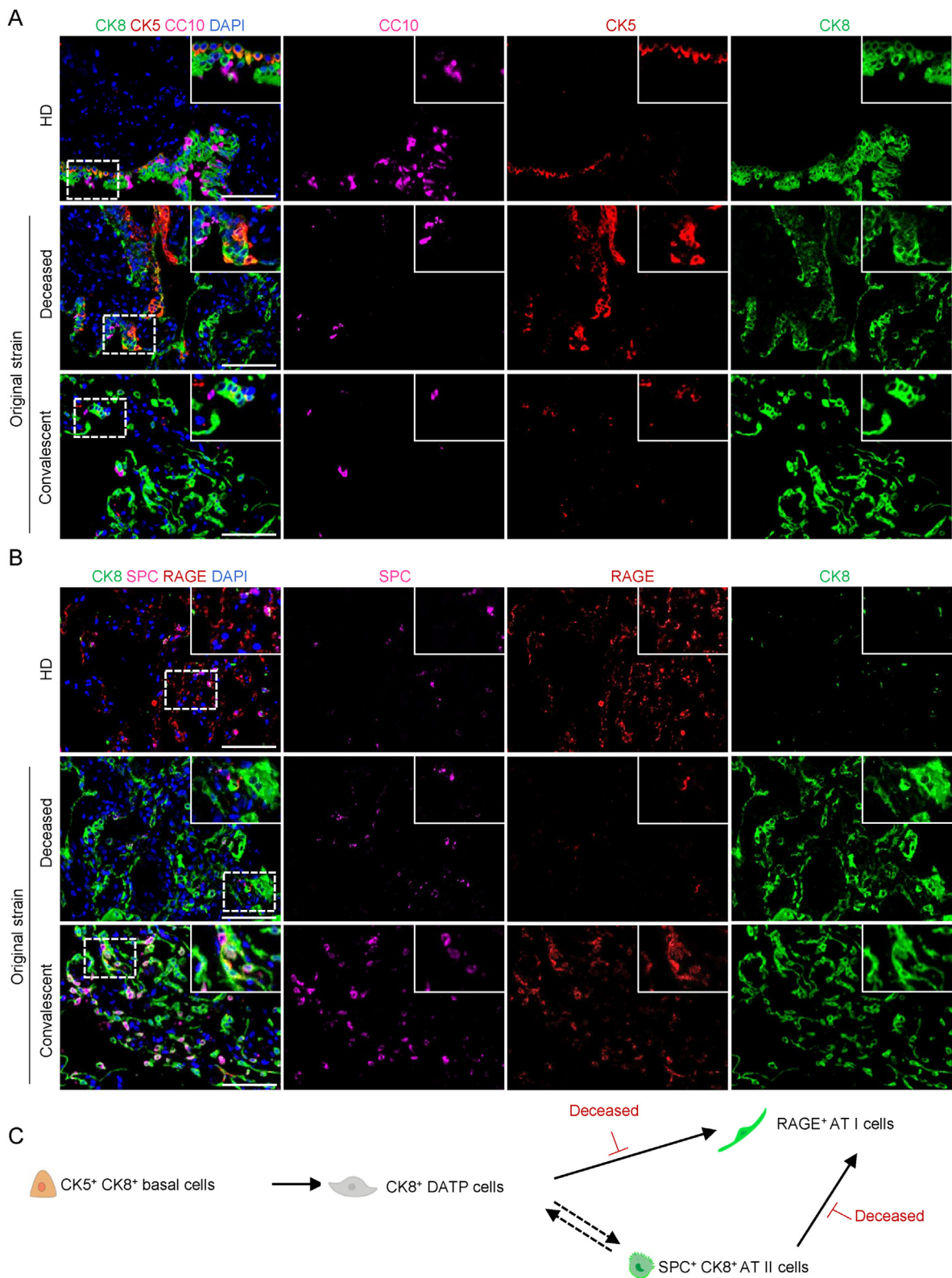


Fig. 2. Abnormal cell proliferation and repair failure in the lungs of the deceased and convalescent COVID-19 patients infected with the original strain. **A** Multi-color IF staining of CK8 (green), CK5 (red) and CC10 (magenta). **B** Multi-color IF staining of CK8 (green), RAGE (red), and SPC (magenta). **C** Schematics of possible cell proliferation and differentiation route in patients with COVID-19. Bars: 200 μ m.

proliferation, which were found in patients who died of SARS-CoV-2 original strain infection, in the two Omicron strain-infected lung samples (Fig. 5D). IF staining of cell markers revealed no obvious changes in the alveolar AT I/II cell numbers or CK8⁺ cell proliferation (Fig. 5F).

4. Discussion

SARS-CoV-2 infection, which varies according to viral strain and individual patient, causes mild flu-like symptoms to severe respiratory

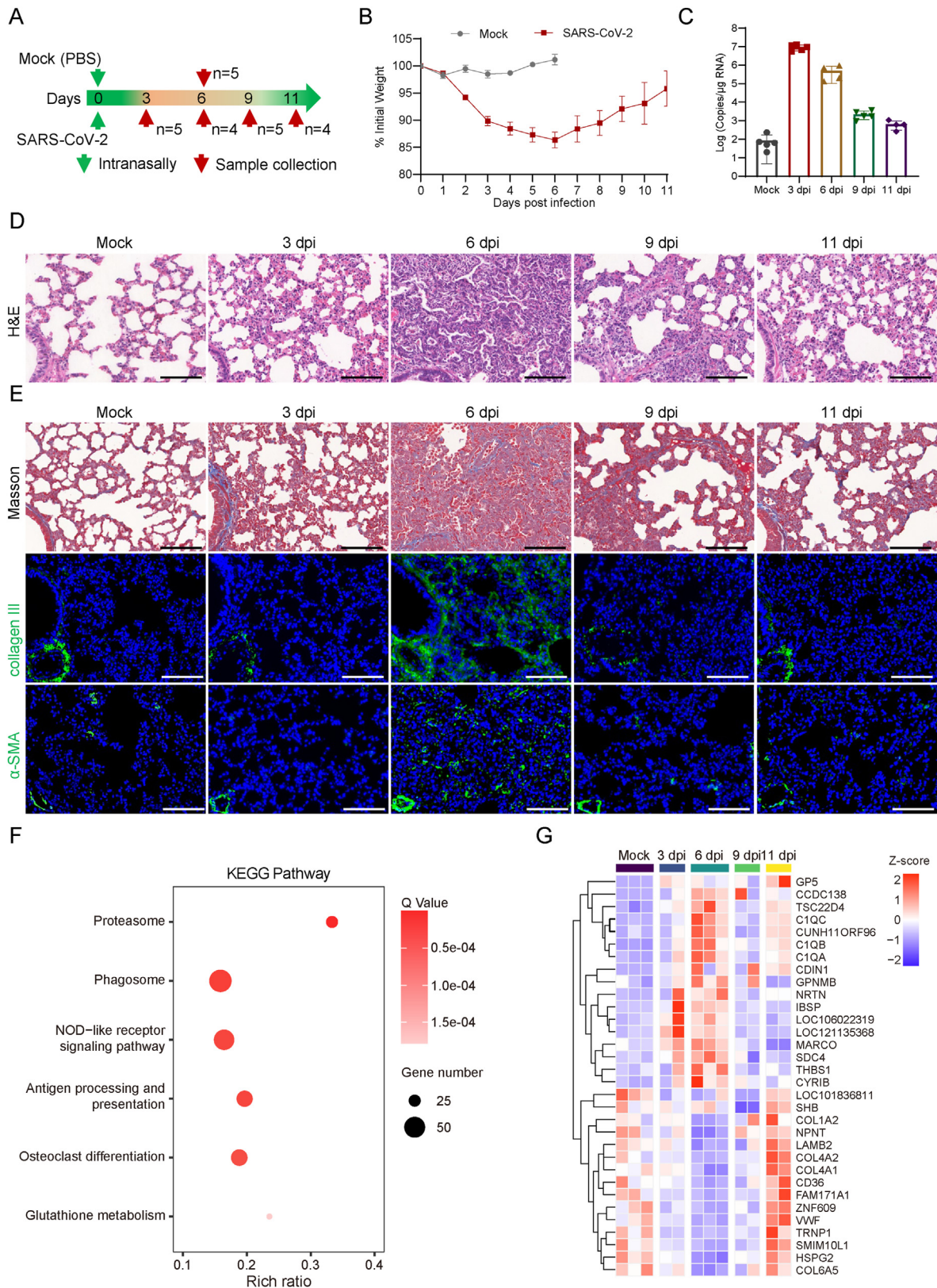


Fig. 3. Pulmonary pathology induced by infection of Syrian golden hamsters with the SARS-CoV-2 original strain. **A** Experimental design. Hamsters were infected with 1×10^5 TCID₅₀ SARS-CoV-2 (50 μ L per hamster) intranasally, and body weight was monitored daily. Lung tissue samples were collected at 3 dpi (n = 5), 6 dpi (n = 4), 9 dpi (n = 5), and 11 dpi (n = 4) for the infection group. The mock infection group (challenged with 50 μ L PBS) samples were collected at 6 dpi (n = 5). **B** Daily body weight changes of hamsters after SARS-CoV-2 infection. Data are presented as mean $\pm$ SD. **C** Viral loads in the lung tissues of hamsters at different dpi were determined using qRT-PCR. Data are presented as mean $\pm$ SD. **D** Hematoxylin and eosin (H&E) staining of lung tissue sections of hamsters at different dpi. **E** Masson's trichrome staining and IF staining of α -SMA and collagen III; bars: 200 μ m. **F**, **G** Transcriptome analyses of lung tissues of hamsters.

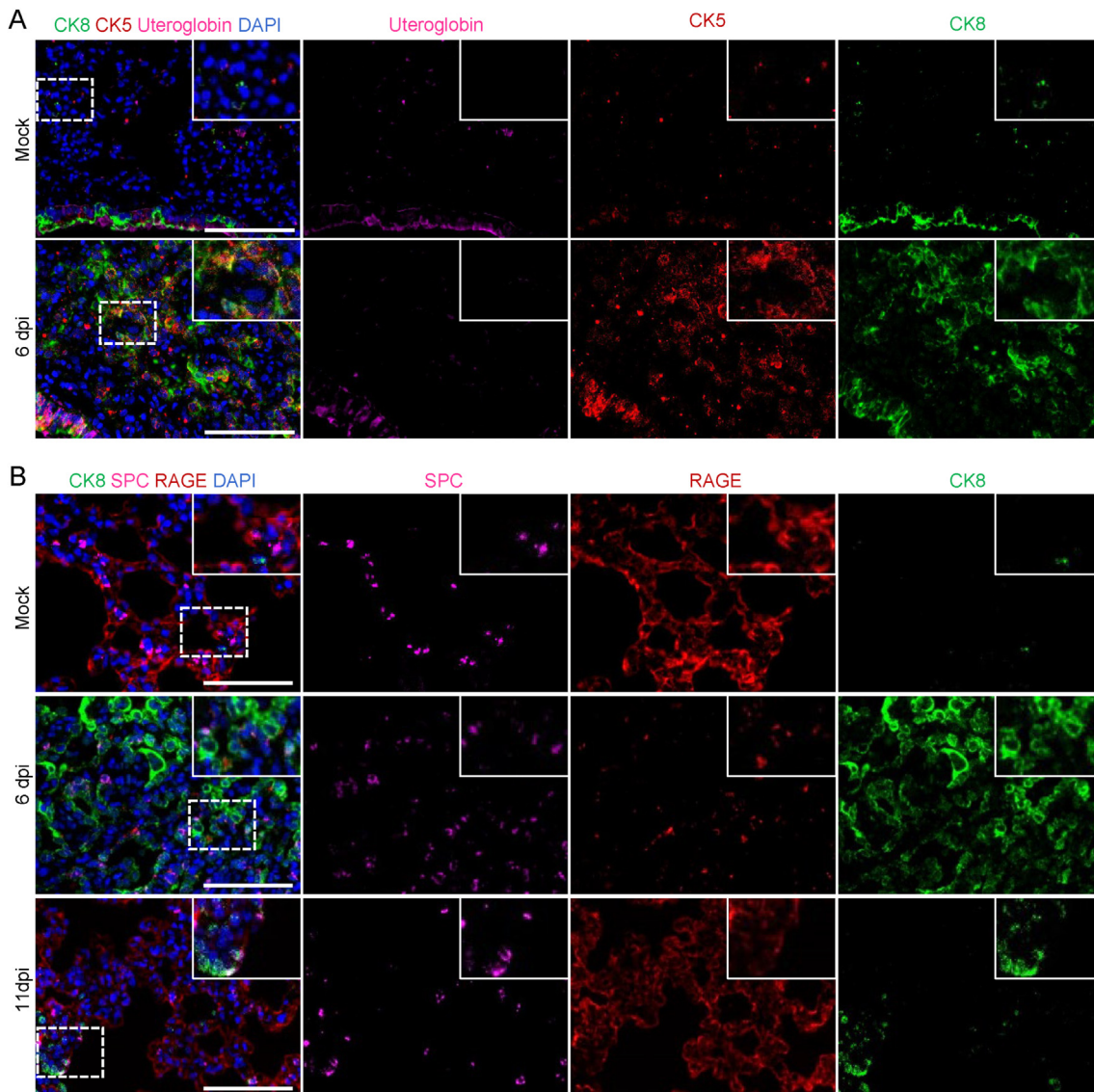


Fig. 4. Abnormal cell proliferation and repairing failure in the lungs of hamsters. **A** Multi-color IF staining of CK8 (green), CK5 (red) and Uteroglobin (magenta) in the lungs of SARS-CoV-2 original strain infected hamsters (6 dpi) or mock infected hamsters. **B** Multi-color IF staining of CK8 (green), RAGE (red) and SPC (magenta) in the lungs of infected hamsters (6 and 11 dpi) or mock infected hamsters. Bars: 200 μ m.

distress syndrome with diffuse lung injury. This study aimed to address the mechanism underlying lung injury remodeling in patients with COVID-19 using patient samples and animal model. In total, 11 deceased COVID-19 patients (nine infected with the original strain and two infected with the Omicron BA.5 strain) and one convalescent patient who was infected with the original strain but died of an unrelated illness after discharge were enrolled. In addition, a self-limiting hamster infection model permissive for natural SARS-CoV-2 infection was established to allow for dynamic study of the lung injury-remodeling process of COVID-19. Comparative studies of these models showed that CK8⁺ DATP cell proliferation is associated with the extent of lung injury and plays a key role in mediating alveolar remodeling in COVID-19.

Analysis of lung injury models induced by bleomycin or influenza virus proved that basal or club cells can differentiate into AT II cells and then differentiate into AT I cells to mediate alveolar repair (Kathiriyai et al., 2020; Zheng et al., 2012). Recently, DATP cells were found to extensively proliferate in alveolar injury models and may have originated from Scgb1a1/CC10⁺ club, CK5⁺, or AT II cells (Ray et al., 2016; Redente et al., 2021; Strunz et al., 2020). We detected proliferating CK5⁺ CK8⁺ basal cells and CK8⁺ DATP cells in the alveoli of both the convalescent

patient with COVID-19 and the deceased COVID-19 patients; these hyperplasia cells lacked the CC10 marker, suggesting that the CK8⁺ cells are originated from CK5⁺ CK8⁺ basal cells in bronchioles and then proliferate in the alveoli (Figs. 1B and 2A). Significant CK8⁺ cell proliferation was also detected in the hamsters infected with SARS-CoV-2, peaking 6 dpi, when alveolar injury was the most severe and a few AT I cells remained (Figs. 3D and 4B, Supplementary Fig. S3B), and thus implying its correlation with disease severity. Proliferative CK5⁺ CK8⁺ cells may also be derived from basal-like cells in hamster bronchioles (Fig. 4A and Supplementary Fig. S4A), which are called lineage-negative epithelial stem/progenitor (LNEP) cells, as has been reported in influenza infection (Vaughan et al., 2015).

CK8⁺ DATP cells may represent intermediate differentiation states of AT II and AT I cells and play a key role in alveolar remodeling and related pulmonary diseases such as idiopathic pulmonary fibrosis (IPF), influenza pneumonia, and COVID-19 (Chen et al., 2020; Choi et al., 2020; Strunz et al., 2020; Wang et al., 2023). In the present study, the proliferating CK8⁺ DATP cells varied according to their differentiation destinations in different cases. In the patients who died of the original SARS-CoV-2 strain, the CK8⁺ DATP cells remained hyperplastic, but

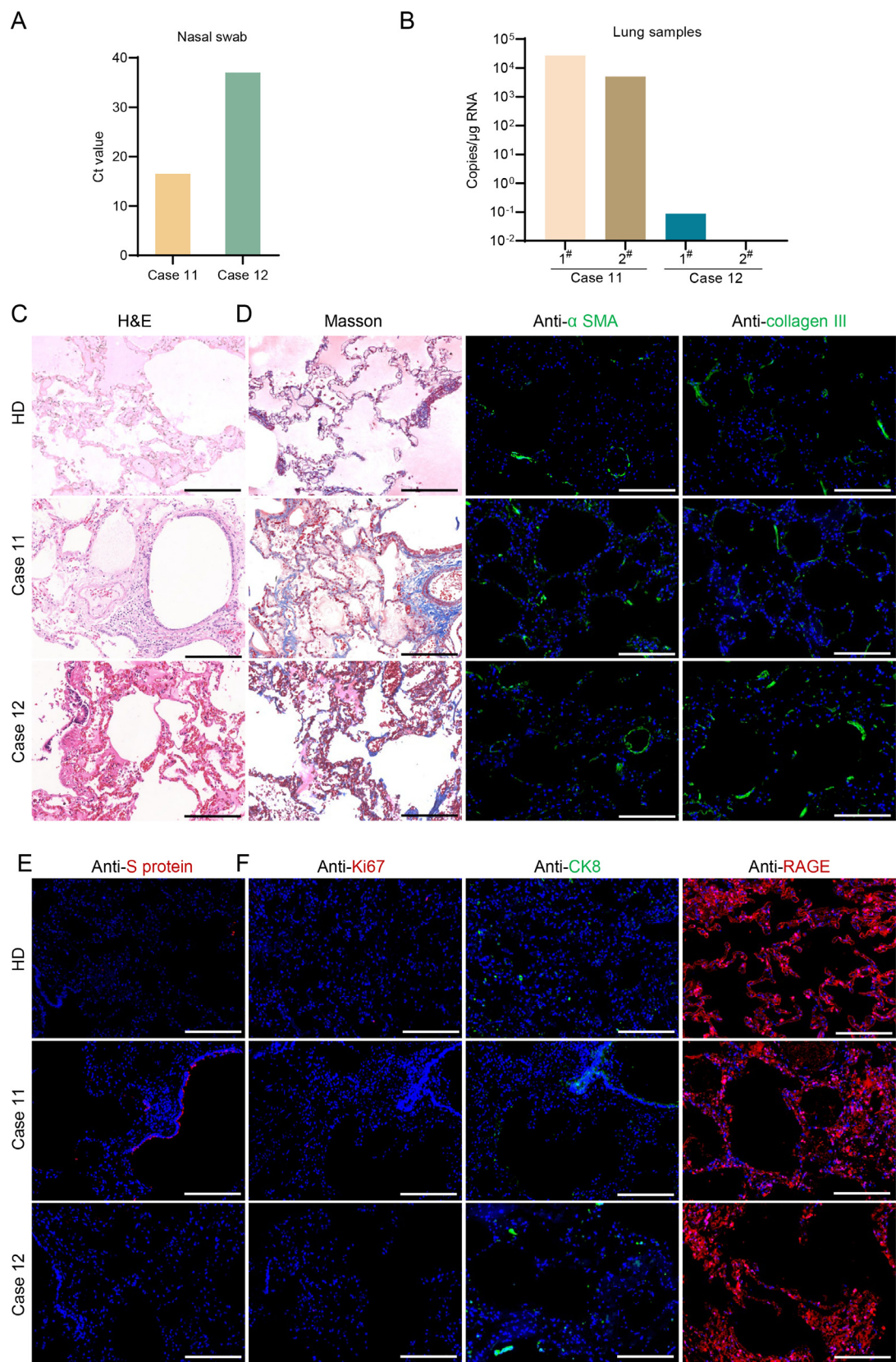


Fig. 5. Histopathological analyses and virus detection of autopsied lungs from COVID-19 patients infected with Omicron BA.5 strain. **A.** Viral detection in the nasal swab. **B.** Viral detection in lungs of two patients infected with Omicron BA.5 strain by qRT-PCR. Lungs #1 and #2 represent two different lung sections in the same patients. **C.** Hematoxylin and eosin (H&E) staining of lung samples of the two patients infected with Omicron BA.5 strain and healthy donor (HD). **D.** Masson's trichrome staining and IF staining of α-SMA and collagen III (green). **E.** IF staining of viral S protein (red). **F.** IF staining of Ki67 (green), CK5 (green), CK8 (green), and RAGE (red). Bars: 200 μm.

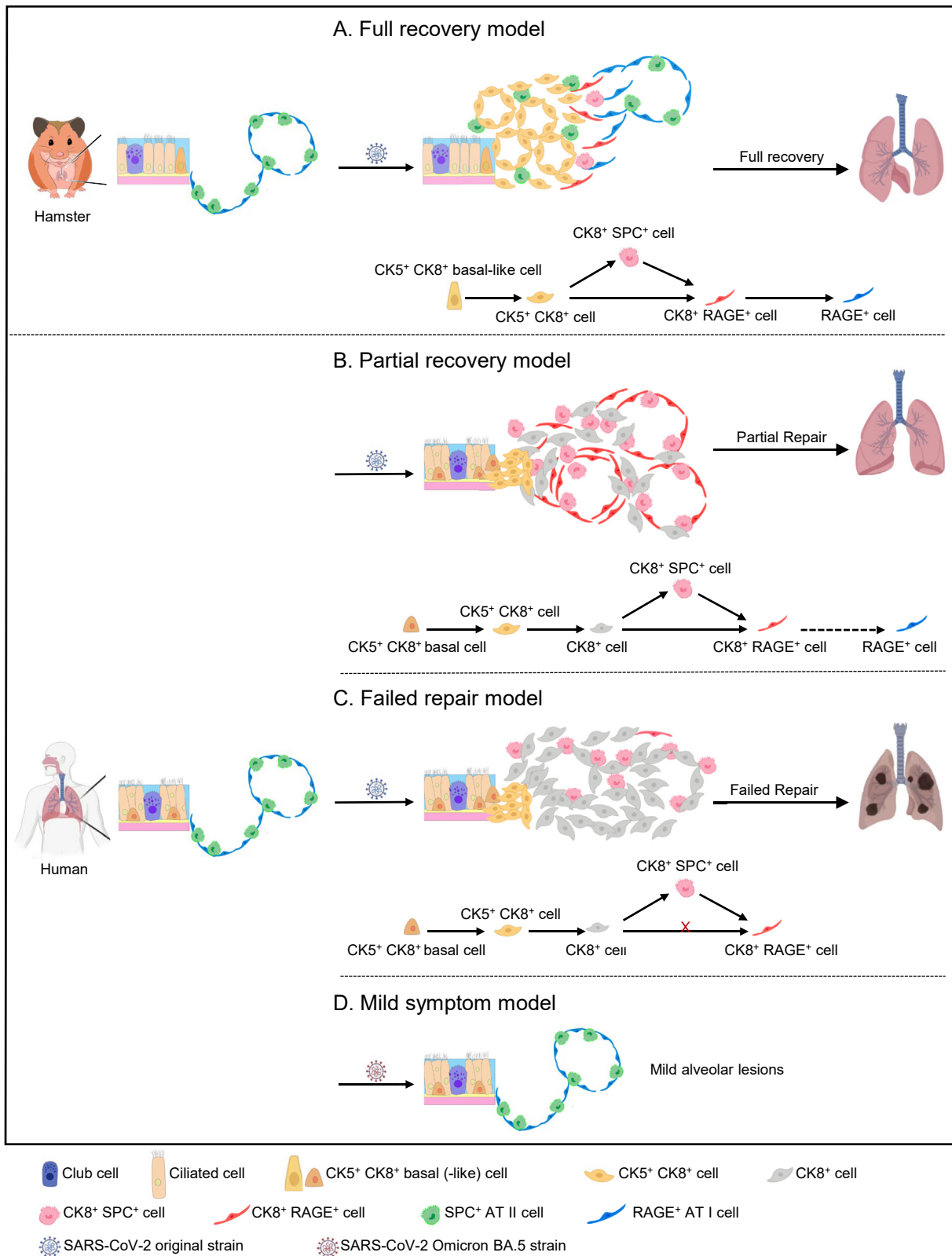


Fig. 6. Possible lung injury-repairing process in different SARS-CoV-2 infection patients and animals. **A** Full recovery model as represented by SARS-CoV-2 infected hamsters. Upon viral infection, bronchiolar CK5⁺ CK8⁺ basal-like cells proliferated significantly either by acquiring the SPC⁺ AT II marker or directly differentiating into RAGE⁺ AT I cells. **B** Partial recovery model represented by a convalescent patient infected with the original SARS-CoV-2 strain. Significant CK8⁺ cell proliferation was detected, which may also be derived from CK5⁺ CK8⁺ basal cells and then differentiate into RAGE⁺ AT I cells as observed in hamsters. However, CK8 marker was still expressed by the newly generated AT I cells. **C** Failed repair model represented by deceased patients infected with SARS-CoV-2 original strain. Severe pulmonary lesions were identified and CK8⁺ cells massively accumulated but most of them lacked the AT I cell marker. **D** Mild symptom model was represented by deceased patients infected with the SARS-CoV-2 Omicron BA.5 strain; they exhibited mild pulmonary lesions, with no CK8⁺ cell proliferation.

few differentiated into AT I cells; while in the convalescent patient with COVID-19, a large portion of CK8⁺ DATP cells co-expressed the AT I marker and transformed into a flattened, elongated AT I cell-like shape, with some of them expressing the SPC marker (AT II cells) (Fig. 2B). The hamster infection model also supported and supplemented the findings based on the human samples. The number of CK8⁺ cells gradually decreased, accompanied by the recovery of AT I cells, from 6 to 11 dpi (Fig. 4B and Supplementary Fig. S3B). A portion of cells co-expressed CK8 and SPC markers (Fig. 4B), suggesting that CK8⁺ cells were upstream of AT I cells during SARS-CoV-2-induced lung remodeling.

In addition to CK8, CK5 appears to be an important regulatory molecule in SARS-CoV-2-induced lung remodeling. In fact, CK5⁺ cells proliferated to the alveoli after mediating alveolar remodeling in an influenza virus infection model (Vaughan et al., 2015). This basal cell marker co-existed with CK8 in bronchioles and migrated to the alveoli in both human samples and the hamster infection model (Figs. 2A and 4A, Supplementary Fig. S3B). We noticed that in samples from both the convalescent patient with COVID-19 and the deceased ones, only a small proportion of CK8⁺ DATP cells expressed CK5 markers simultaneously (Fig. 2A), whereas most proliferating cells co-expressed CK8 and CK5 markers in hamsters throughout the infection process (Fig. 4A and Supplementary Fig. S4A). Another difference between the hamster infection model and patients was that the lungs of hamsters could rapidly remodel post SARS-CoV-2 infection (severely damaged at 6 dpi and fully recovered at 11 dpi) (Fig. 3D). Interestingly, most CK8⁺ DATP hamster cells expressed the proliferation marker Ki67 (Supplementary Fig. S4B). In contrast, in human patients, although CK8⁺ cells were highly proliferative, the ratio of CK8⁺ Ki67⁺ DATP cells was low (Supplementary Fig. S2B). This may partially explain the rapid lung remodeling in the hamsters compared to the humans we studied.

After alveolar injury, successful alveolar remodeling is accompanied by the successful differentiation of AT I cells. CK8⁺ cells, currently reported as an intermediate form of differentiation for SPC⁺ AT II and RAGE⁺ AT I cells, are crucial for alveolar remodeling (Strunz et al., 2020). We observed that the accumulation of CK8⁺ cells in the lungs of patients with COVID-19 and their unsuccessful differentiation into AT I cells may be related to pulmonary fibrosis caused by SARS-CoV-2 infection. Indeed, transcriptomic analysis identified significant upregulation of ECM-related pathways in deceased patients with COVID-19 compared to healthy donors (Fig. 1C and D). Although the regeneration of RAGE⁺ AT I cells was successful in the convalescent patient, the CK8 marker was still abundant, indicating that the alveolar epithelium had not fully recovered, which may be related to pulmonary sequelae such as respiratory distress caused by COVID-19. However, the precise mechanisms underlying the differentiation of AT II or DATP cells into AT I cells is unclear. P53, TGF- β , abnormal Hippo, and Wnt pathways are associated with alveolar remodeling in acute lung injury models or progressive lung fibrosis (Finn et al., 2019; Kaiser et al., 2023; Liu et al., 2016; Wu et al., 2020; Zacharias et al., 2018). For example, P53 and Wnt pathways are involved in the proliferation of mesenchymal stem cells (Kaiser et al., 2023; Zacharias et al., 2018; Zuo et al., 2015). TGF- β is related to the differentiation of AT II cells into AT I cells and is activated in DATP cells for fibrosis progression (Kobayashi et al., 2020; Wang et al., 2023; Wu et al., 2020). YAP/TAZ in the Hippo pathway are crucial for mesenchymal stem cells differentiation into RAGE⁺ AT I cells (Liu et al., 2016). In addition, how bronchial DATP cells migrate into alveoli remains enigmatic. During embryonic lung development, alveolar progenitor cells undergo an actin-based cell protrusion from airway epithelium toward the mesenchyme by FGF10-mediated chemotaxis (Li et al., 2018). Further in-depth analysis of transcriptome data between patients and hamsters may facilitate to unravel the differentiation and migration processes and the detailed regulatory mechanism of DATP cell, and elucidate alveolar injury and the remodeling mechanism of COVID-19. In addition to studying the original strain infection, two autopsied lungs infected with the Omicron BA.5 variant were analyzed, and only mild lung lesions were identified in both cases (Fig. 5C). Diffuse alveolar

damage in Omicron BA.1/BA.2 cases is considerably lower than that in the case of the original strain infection (Märkl et al., 2023; Schwab et al., 2023). No CK8⁺ DATP cell proliferation or pulmonary fibrosis was detected in either patient (Fig. 5D and F), although viral RNA was detected in the nasal swabs and lungs (Fig. 5A, B, and E). These results indicated a specific relationship between the degree of lung injury and CK8 hyperplasia in patients with COVID-19. However, only two autopsied Omicron cases were included in this study, and additional cases may provide more comprehensive information.

In summary, we identified the pulmonary cell differentiation routes following SARS-CoV-2 infection in patients or hamster model (Fig. 6). In hamsters infected with the original strain (full recovery model), CK5⁺ CK8⁺ basal-like cells proliferated, either by obtaining the SPC⁺ AT II marker or by directly differentiating into RAGE⁺ AT I cells (Fig. 6A). In the convalescent patient infected with the original strain (partial recovery model), CK8⁺ cells might be derived from CK5⁺ CK8⁺ basal cells and then differentiate into RAGE⁺ AT I cells, as observed in infected hamsters (Fig. 6B). However, CK8 coexists with most AT I cells. In deceased patients infected with the original strain (failed repair model), severe pulmonary lesions were identified, and CK8⁺ cells accumulated considerably; however, most of them lacked AT I cell markers (Fig. 6C). In deceased patients infected with the Omicron strain (mild symptom model), mild pulmonary lesions with no CK8⁺ cell proliferation were observed (Fig. 6D).

These findings indicate that CK8⁺ cells could successfully differentiate into RAGE⁺ AT I cells, which are key for alveolar remodeling after SARS-CoV-2 infection. The blocked differentiation of CK8⁺ cells may be related to pulmonary fibrosis caused by SARS-CoV-2 infection. Clinical experiments on stem cells are ongoing in patients with severe acute respiratory distress syndrome and chronic obstructive pulmonary disease (COPD) (Simonson et al., 2015; Weiss et al., 2013). Similarly, our data suggest that the differentiation of CK8⁺ mesenchymal stem cells or supplementation with normal mesenchymal stem cells for remodeling the alveolar epithelium and reducing pulmonary sequelae may be an optimal strategy for the treatment of severe lung injury caused by SARS-CoV-2 infection.

5. Conclusions

By systematic analyses, the lungs of nine patients who died of COVID-19 in early 2020 showed varying extents of CK8⁺ DATP cell proliferation, which derived from bronchial CK5⁺ basal cells, but ultimately failed to differentiate into AT I cells for proper lung remodeling. While in a convalescent COVID-19 patient who died of a different illness, accumulated CK8⁺ cells were undergoing transition into AT I cells. By exploring self-limiting hamster infection model, the proliferation and regression of CK8⁺ cells were closely related to the disease course. By contrast, two patients infected with Omicron variant exhibited mild pulmonary injury with no CK8⁺ cell proliferation. These results suggest a critical role of CK8⁺ DATP cells in mediating alveolar remodeling upon SARS-CoV-2 infection and their potentials as diagnostic markers and therapeutic targets.

Data availability

The RNA-seq data supporting the findings of this study are available at CNGB Sequence Archive (CNSA) of China National GeneBank Data-Base (CNGBdb) with accession number CNP0004841 and CNP0004790.

Ethics statement

Lung autopsy samples in this study were provided by the ethics Committee at Wuhan Jinyintan Hospital Ethics Committee (permission number: KY-2020-15.01) and Tongji medical school, Huazhong University of Science and Technology (permission number: [2021] IEC-A001). Healthy donors' ethics were provided by the ethics Committee at Tongji medical school, Huazhong University of Science and Technology

(permission number: 2022-S099). Written informed consent was obtained from the patient's family before autopsy. Animal experiment in this study was approved by the Institutional Animal Care and Use Committee (IACUC) of Wuhan institute of Virology, Chinese Academy of Sciences (ethics number: WIVA01202001).

Author contributions

Yufeng Li: methodology, validation, investigation, data curation, formal analysis, visualization, writing-original draft, writing reviewing, and editing. Hengrui Hu: methodology, formal analysis, data curation. Jia Liu: methodology, investigation, supervision. Longda Ma: methodology, investigation. Xi Wang: investigation, writing-reviewing. Liang Liu: resource, funding acquisition. Qian Liu: resource, funding acquisition. Liang Ren: resource. Jiang Li: project administration, investigation. Fei Deng: resource, investigation. Zhihong Hu: conceptualization, supervision, writing-reviewing. Yiwu Zhou: conceptualization, supervision, writing-reviewing. Manli Wang: conceptualization, supervision, funding acquisition, writing-reviewing, and editing.

Conflict of interest

Prof. Fei Deng is an editorial board member for *Virologica Sinica* and was not involved in the editorial review or the decision to publish this article. All the authors declare no conflicts of interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.virs.2024.03.007>.

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